Vignesh Subramanian

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Research Interests

I am a Ph.D. student at Georgia Tech working on building reinforcement learning systems that can both generalize reliably and provide clear, verifiable guarantees about their behavior. My research combines insights from formal methods, neuro-symbolic reasoning, and large language models to design agents that not only learn effectively but can also justify and certify their decisions in safety-critical settings.

Education

Georgia Institute of Technology — **Aug 2023 – Ongoing**
Ph.D. in Computer Science | 3rd Year | GPA: 3.9 — Atlanta, Georgia
Advisor – Prof. Suguman Bansal | *Anticipated Graduation: May 2028*

National Institute of Technology, Tiruchirappalli — **Aug 2019 - April 2023**
B.Tech. - Electrical and Electronics Engineering — Tiruchirappalli, Tamil Nadu
Minor - Computer Science Engineering

Relevant Coursework: Logic in Computer Science, SAT/SMT solvers, Machine Learning and Deep Learning, Software Analysis and Testing, Artificial Intelligence, Data Structures and Algorithms

Publications

• Conferences

Inductive Generalization in Reinforcement Learning from Specifications
Automated Technology for Verification and Analysis, 2025
Vignesh Subramanian, Rohit Kushwah, Subhajit Roy, Suguman Bansal **[Link]**

Certification-Guided Evaluation of Reinforcement Learning Generalization (Presentation Only)
Symposium on AI Verification, 2025
Vignesh Subramanian, Djordje Zikelic, Suguman Bansal **[Link]**

Reinforcement Learning for Stochastic Max-Plus Linear Systems
Conference on Decision and Control, IEEE 2023
Vignesh Subramanian, Farzaneh Farhadi, Sadegh Soudjani **[Link]**

A Novel Facial Emotion Recognition Model Using Segmentation VGG-19 Architecture
International Journal of Information Technology, Springer, 2023
Vignesh Subramanian, Savithadevi, M. Sridevi, Rajeswari Sridhar **[Link]**

SIHeDA-Net: Sensor to Image Heterogeneous Domain Adaptation for Sign Language Detection
Medical Imaging with Deep Learning, 2022
Ishikaa Lunawat, **Vignesh Subramanian**, S. P. Sharan **[Link]**

• Workshops

Certification-Guided Evaluation of Reinforcement Learning Generalization (Short Paper)
AAAI - Workshop on Generalization in Planning, 2025
Vignesh Subramanian, Djordje Zikelic, Suguman Bansal **[Link]**

Inductive Generalization in Reinforcement Learning from Specifications (Short Paper)
NeurIPS - Workshop on Generalization in Planning, 2023
Rohit Kushwah, **Vignesh Subramanian**, Subhajit Roy, Suguman Bansal **[Link]**

Teaching Experience

CS 8803 - SAT/SMT Solvers **Fall 2024**
Georgia Institute of Technology *Atlanta, Georgia*

- Guiding students on advanced topics in SAT and SMT solvers, including model checking, symbolic execution, neural network verification, and formal verification techniques.
- Supporting assignments and projects that integrate LLM-based reinforcement learning, LLM inference, and proof synthesis for solving complex problems in solver architecture and proof complexity.

CS 8803 - Formal Methods in Reinforcement Learning **Fall 2025**
Georgia Institute of Technology *Atlanta, Georgia*

- Assisting instruction and research-oriented discussions covering formal verification, neuro-symbolic reasoning, safe RL, correctness guarantees, and specification-guided learning.
- Supporting student final projects, literature reviews, and research direction selection at the intersection of reinforcement learning, formal methods, and trustworthy AI.

Internships

Newcastle University, United Kingdom | Paper link **Jan 2022 - Nov 2022**
Research Intern - Mentor: Prof. Sadegh Soudjani *Newcastle, England*

- Proposed a novel optimization strategy using Deep Q-Learning for Stochastic Max-Plus-Linear Discrete Event Systems under uncertainties, achieving a 2.5x speedup over Model Predictive Control for minimizing stochastic delays in railway systems, published at **Conference on Decision and Control, IEEE, 2023**.

Samsung R&D Institute, Bangalore | Report link **Aug 2021 - Feb 2022**
Research Intern - Mentor: Mr. Sujoy Saha, Mr. Rajat Kumar Jain *Karnataka, India*

- Developed a Tri-Stage Occlusion Handling Normal Map Estimation Algorithm to enhance 3D human normal map estimation on blurred, noisy, low-res 2D images, achieving 92.78% IoU (vs. 71.26% baseline) by effectively addressing occlusions from shadows, blur, and background.

National Institute of Technology, Tiruchirappalli | Paper link | Github link **Feb 2021 - Dec 2021**
Research Intern - Mentor: Prof. M. Sridevi *Tamil Nadu, India*

- Designed a novel CNN architecture integrating U-Net and VGG layers for Facial Emotion Recognition, achieving 75.97% accuracy on the FER-2013 dataset, ranking in the **Top 5** on the global leaderboard. Findings were published in the **International Journal of Information Technology, Springer**.

Technical Skills

- **Languages**: C/C++, Python, Java, JavaScript, Lean
- **Packages/Frameworks**: PyTorch, Tensorflow, Keras, Scikit-learn, OpenCV, OpenAI Gym, Stable Baselines
- **Reinforcement Learning**: Policy Gradient Methods, Generalization Algorithms, Safety Certificates for RL, Reachability and Safety Analysis
- **Formal Verification**: Model Checking, Formal Methods for Safety Verification
- **LLM**: Pre-training and fine-tuning large-scale transformer models, LLM-based inference for complex logical queries, reinforcement learning from human feedback (RLHF), and automated proof generation.

Positions Of Responsibility

Reviewer **2025**
AAAI — Reviewer for the AAAI Workshop on Generalization in Planning.

Student Research Mentor, Machine Learning and AI **2020 – 2023**
Spider R&D, NIT Trichy — Led ML/AI projects; built SIHeDA-NET (Paper) and GISiL (Demo).

Coordinator, Workshops and Publicity **2020 – 2021**
Currents Symposium, NIT Trichy — Managed workshops, publicity; hosted 600+ participants.